

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform the traditional loan approval process using machine learning, boosting operational efficiency and accuracy. It tackles manual system inefficiencies by replacing slow human underwriting with instant, ML-driven predictions. Key features include a highly accurate XGBoost credit model, robust data preprocessing, and a seamless real-time decision-making web interface built on Flask.

Project Overview	
Objective	The primary objective is to revolutionize the loan approval process by implementing an advanced XGBoost machine learning model integrated with a Flask web application, ensuring instantaneous, unbiased, and accurate credit assessments.
Scope	The project comprehensively addresses the entire machine learning lifecycle—from handling missing values and balancing data with SMOTE to model training—and deploys it into a fully accessible, responsive web platform for real-time predictions.
Problem Statement	
Description	The manual verification of complex applicant financial factors (such as combined incomes, dependents, and property area) is highly subjective, prone to human error, and takes far too much time, adversely affecting operational efficiency and customer satisfaction.
Impact	Solving these issues with automation will result in instantaneous processing, the elimination of human bias, reduced default risks for the bank, and a dramatically improved, stress-free experience for loan applicants.
Proposed Solution	
Approach	Employing advanced machine learning techniques (specifically XGBoost and SMOTE data balancing) to analyze and predict applicant creditworthiness based on 11 distinct financial metrics, safely deployed via a lightweight Flask backend.
Key Features	- Implementation of a machine learning-based credit assessment model.

	- Interactive, real-time decision-making web portal with a Premium Glassmorphism UI - Automated dynamic risk banding (calculating exact probabilities for Low, Moderate, or High risk).
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard x86 CPU
Memory	RAM specifications	4GB to 8 GB
Storage	Disk space for data, models, and logs	50 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit learn, xgboost, pandas, numpy, matplotlib, seaborn, imbalanced-learn (SMOTE)
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle Loan Prediction dataset, 614 records, CSV format